

V_{ud} project notes

1 Beta functions

1.1 Beta functions defined in Arason

The beta functions at NLO for EW theory before symmetry breaking are defined as:

$$\begin{aligned}\beta_{g_1} &= \frac{41}{10}g_1(t)^3 \\ \beta_{g_2} &= -\frac{19}{6}g_2(t)^3 \\ \beta_{y_t} &= y_t(t) \left(\frac{9}{2}y_t(t)^2 - \left(\frac{17}{20}g_1(t)^2 + \frac{9}{4}g_2(t)^2 \right) \right) \\ \beta_\lambda &= 12\lambda(t)^2 - \lambda(t) \left(\frac{9}{5}g_1(t)^2 + 9g_2(t)^2 \right) + \frac{9}{4} \left(\frac{3}{25}g_1(t)^4 + \frac{2}{5}g_1(t)^2g_2(t)^2 + g_2(t)^4 \right) \\ &\quad + 12\lambda(t)y_t(t)^2 - 12y_t(t)^4\end{aligned}$$

with $SU(2) \times U(1)_Y$ and $t = \ln(\mu)$.

The running of the fundamental parameters is given by

$$\begin{aligned}\frac{dg_1(t)}{dt} &= \frac{1}{16\pi^2}\beta_{g_1} \\ \frac{dg_2(t)}{dt} &= \frac{1}{16\pi^2}\beta_{g_2} \\ \frac{dy_t(t)}{dt} &= \frac{1}{16\pi^2}\beta_{y_t} \\ \frac{d\lambda(t)}{dt} &= \frac{1}{16\pi^2}\beta_\lambda\end{aligned}$$